

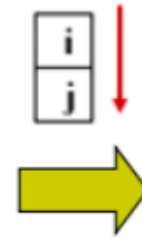
ឧទាហរណ៍ លេខសម្គាល់ 66172110157-8



Example of Vertical GLCM

0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3

Vertical Co-occurrence



	0	1	2	3
0	3	0	2	0
1	0	2	2	0
2	0	0	1	2
3	0	0	0	0

Count Matrix

3	0	0	0
0	2	0	0
2	2	1	0
0	0	2	0

Transpose Matrix



Example of Vertical GLCM(ต่อ)



8	6	0	2	0
6	0	4	2	0
8	2	2	2	2
2	0	0	2	0

Symmetrical Matrix

↓

$$24 \times \frac{1}{24}$$

Vertical GLCM (Probability)

0.25	0	0.083	0
0	0.167	0.083	0
0.083	0.083	0.083	0.083
0	0	0.083	0

Contrast weight

0	1	4	9
1	0	1	4
4	1	0	1
9	4	1	0

Vertical GLCM (Probability)

0.25	0	0.083	0
0	0.167	0.083	0
0.083	0.083	0.083	0.083
0	0	0.083	0

*

Contrast Matrix

0	0	0.332	0
0	0	0.083	0
0.332	0.083	0	0.083
0	0	0.083	0

$$= 0.332 + 0.083 + 0.332 + 0.083 + 0.083 + 0.083$$

$$\rightarrow = 1$$

=

similarity weight

1	0.5	0.333	0.25
0.5	1	0.5	0.333
0.333	0.5	1	0.5
0.25	0.333	0.5	1

Vertical GLCM (Probability)

0.25	0	0.083	0
0	0.167	0.083	0
0.083	0.083	0.083	0.083
0	0	0.083	0

*

Similarity Matrix

=

0.25	0	0.028	0
0	0.167	0.041	0
0.028	0.041	0.083	0.041
0	0	0.041	0

$$= 0.25 + 0.028 + 0.167 + 0.041 + 0.028 + 0.041 + 0.083 + 0.041 + 0.041$$

$$= 0.72$$

Dissimilarity weight

0	1	2	3
1	0	1	2
2	1	0	1
3	2	1	0

Vertical GLCM (Probability)

0.25	0	0.083	0
0	0.167	0.083	0
0.083	0.083	0.083	0.083
0	0	0.083	0

~~*~~

Dissimilarity Matrix

0	0	0.166	0
0	0	0.083	0
0.166	0.083	0	0.083
0	0	0.083	0

$$= 0.166 + 0.083 + 0.166 + 0.083 + 0.083 + 0.083$$

$$= 0.664$$

//

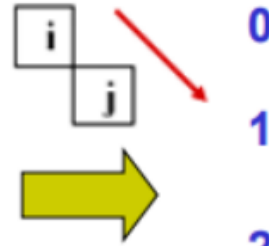


Example of Diagonal GLCM



0	0	1	1
0	0	1	1
0	2	2	2
2	2	3	3

Diagonal Co-occurrence



	0	1	2	3
0	1	1	3	0
1	0	1	2	0
2	0	0	0	2
3	0	0	0	0

Count Matrix

1	0	0	0
1	1	0	0
3	1	0	0
0	0	2	0

Transpose Matrix



Example of Diagonal GLCM(ต่อ)



78 ←

6	2	1	3	0
4	1	2	1	0
6	3	1	0	2
2	0	0	2	0

Symmetrical Matrix

$$\frac{1}{18} \times$$

Diagonal GLCM (Probability)

0.111	0.055	0.166	0
0.055	0.111	0.055	0
0.166	0.055	0	0.111
0	0	0.111	0

Contrast weight

0	1	4	9
1	0	1	4
4	1	0	1
9	4	1	0



Diagonal GLCM (Probability)

0.111	0.055	0.166	0
0.055	0.111	0.055	0
0.166	0.055	0	0.111
0	0	0.111	0

Contrast Matrix

0	0.055	0.664	0
0.055	0	0.055	0
0.664	0.055	0	0.111
0	0	0.111	0

||



$$= 0.055 + 0.664 + 0.055 + 0.055 + 0.664 + 0.055 \\ + 0.111 + 0.111$$

$$= 1.77$$

similarity weight

1	0.5	0.333	0.25
0.5	1	0.5	0.333
0.333	0.5	1	0.5
0.25	0.333	0.5	1



Diagonal GLCM (Probability)

0.111	0.055	0.166	0
0.055	0.111	0.055	0
0.166	0.055	0	0.111
0	0	0.111	0

Similarity Matrix

0.111	0.027	0.055	0
0.027	0.111	0.027	0
0.055	0.027	0	0.055
0	0	0.055	0

==

$$\begin{aligned}
 &= 0.111 + 0.027 + 0.055 + 0.027 + 0.111 \\
 &\quad + 0.027 + 0.055 + 0.027 + 0.055 + 0.055 \\
 &= 0.55
 \end{aligned}$$

Dissimilarity weight

0	1	2	3
1	0	1	2
2	1	0	1
3	2	1	0

Diagonal GLCM (Probability)

0.111	0.055	0.166	0
0.055	0.111	0.055	0
0.166	0.055	0	0.111
0	0	0.111	0



Dissimilarity Matrix

0	0.055	0.332	0
0.055	0	0.055	0
0.332	0.055	0	0.111
0	0	0.111	0

||



$$= 0.055 + 0.332 + 0.055 + 0.055 + 0.332 + 0.055$$

$$+ 0.111 + 0.111$$

$$= 1.106$$